

Cultural Homophily in Grand Slam Doubles

ATP Men's Grand Slam Doubles, 2018–2025 · 1,949 matches after dropping 48 retirements / walkovers · Regression sample: 3,744 team-obs (complete ranking, nationality/language for all 4 players; GS experience imputed to 0 for first-timers)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family).

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average GS doubles appearances (*exp_mean*), and its square. Culture measures enter one at a time. Rank gap between teammates was tested as an additional control and is insignificant throughout; it is excluded from all main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped dataset (2018–2025, incl. AO/RG/USO 2020; Wimbledon 2020 cancelled)	1,997	—	3,994
2	Drop retirements / walkovers (20 walkovers · 10 retired in set 1 · 18 retired in set 2+)	1,949	−48	3,898
3	Drop: doubles ranking incomplete for ≥1 player	1,935	−14	3,870
4	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, <code>manual_nationality.csv</code>)	1,935	0	3,870
5	Exclude Olympic matches (Tokyo 2021 · Paris 2024)	1,872	−63	3,744
6	GS experience: 216 obs are first-time GS participants (no prior appearances); <i>exp_mean</i> imputed to 0 — no obs dropped (these teams have mean rank ≈193 vs ≈124 for the rest; dropping would create selection bias)	—	0	3,744
→ Grand Slams regression sample (Tables 3–4)		—		3,744

Table 4 (tiebreak win): 1,148 tiebreaks × 2 = 2,296 team-obs (919 unique matches).

3. Pressure Outcome Counts

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	476	24.4%
Set-2 tiebreak (7-pt)	496	25.4%
Any regular tiebreak (sets 1–2)	858	44.0%
Match tiebreak / super-tb (10-pt, set 3)	24	1.2%
Any tiebreak (all types)	958	49.2%
Match went to 3 sets	852	43.7%
Comeback wins (winner lost set 1)	384	19.7%

All matches (GS + Olympics, N=1,949). Match tiebreaks concentrated in Olympics (N=20) and Wimbledon post-2018 (N=4).

4. Olympic Cycle Descriptive Evidence

Table 2A — Tokyo 2021 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Tokyo	AO18 – USO19	1,006	40.1%	59.9%	54.8%	45.2%	55.7%	44.3%
Tokyo Prep	AO20 – Wim21 (ex Wim20)	676	40.4%	59.6%	55.0%	45.0%	55.2%	44.8%
Post-Tokyo	USO21 – USO22	604	39.2%	60.8%	55.0%	45.0%	55.8%	44.2%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Table 2B — Paris 2024 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Paris	AO22 – USO22	486	39.5%	60.5%	54.3%	45.7%	55.3%	44.7%
Paris Prep	AO23 – Wim24	860	42.6%	57.4%	57.6%	42.4%	58.0%	42.0%
Post-Paris	USO24 – USO25	626	43.9%	56.1%	58.8%	41.2%	60.2%	39.8%

N teams = GS matches × 2.

Cultural Homophily in Grand Slam Doubles - Tokyo 2021 Cycle
(Same vs. Diff, by Tournament-Year)

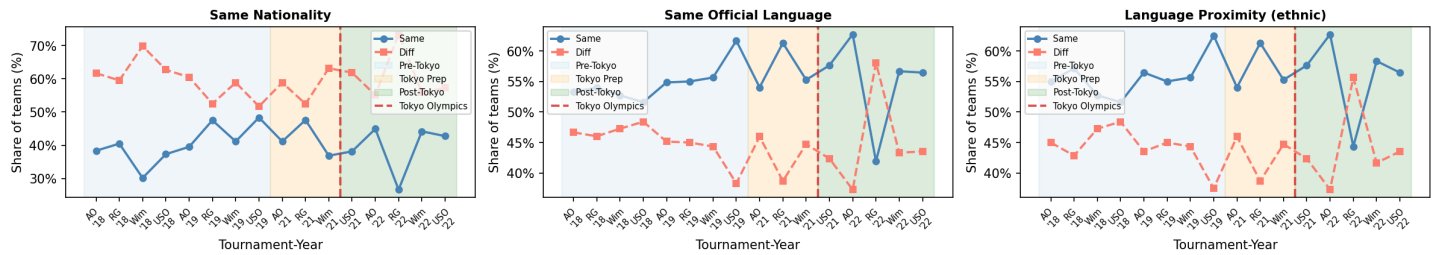


Figure 1. Share of same- vs. different-culture teams by tournament-year, Grand Slams only. Shaded: Pre-Tokyo (blue), Tokyo Prep (orange), Post-Tokyo (green). Red dashed = Tokyo Olympics 2021.

Cultural Homophily in Grand Slam Doubles - Paris 2024 Cycle
(Same vs. Diff, by Tournament-Year)

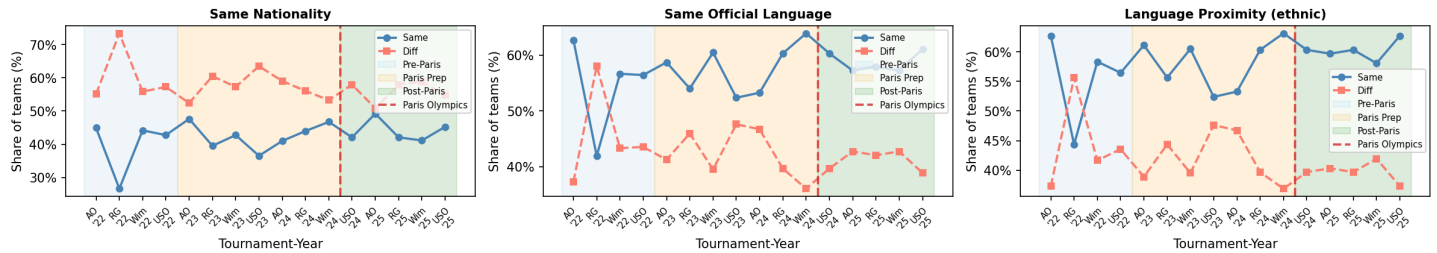


Figure 2. Paris 2024 cycle. Red dashed = Paris Olympics 2024.

Cultural Homophily in Grand Slam Doubles -- Full Timeline 2018-2025
(Tokyo & Paris Olympic Cycles Combined, Same vs. Diff)

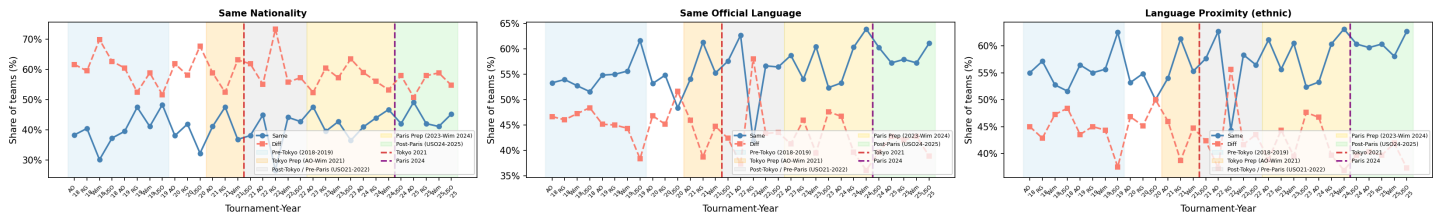


Figure 3. Both Olympic cycles on a single timeline (2018–2025). Vertical dashed lines mark Tokyo 2021 and Paris 2024.

5. Regression Results

Average marginal effects (AME, dP/dx) from logit models. Each column is a separate model. Standard errors in parentheses, clustered by match. FE: tournament $\times$ year, round. All specifications include GS experience controls (*exp_mean* and its square). Culture measures enter one at a time.

Table 3 — Outcome: Match Win | N = 3,744 team-obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0368** (0.0163) p=0.024	+0.0460*** (0.0156) p=0.003	+0.0426*** (0.0156) p=0.006
Team avg. doubles rank	−0.0007*** (0.0001)	−0.0007*** (0.0001)	−0.0007*** (0.0001)
Opponent avg. doubles rank	+0.0007*** (0.0001)	+0.0007*** (0.0001)	+0.0007*** (0.0001)
Top-100 singles player	−0.0050 (0.0318) p=0.874	−0.0038 (0.0318) p=0.906	−0.0032 (0.0318) p=0.920
Team avg GS appearances	+0.0200*** (0.0051) p=0.000	+0.0197*** (0.0051) p=0.000	+0.0195*** (0.0051) p=0.000
GS experience squared	−0.0007*** (0.0002) p=0.002	−0.0007*** (0.0002) p=0.002	−0.0007*** (0.0002) p=0.002
Pseudo-R ²	0.096	0.096	0.096

*** p<0.01 ** p<0.05 * p<0.10. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant at the 5% level (nationality p=0.024, language p=0.003, proximity p=0.006). Experience shows a concave effect: positive linear (p<0.01), negative quadratic (p<0.01). Top-100 singles indicator is not significant: within-GS doubles, having a top singles player does not predict match win once ranking and experience are controlled.

Table 4 — Outcome: Tiebreak Win | Unit: team × tiebreak | N = 2,296 obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0267 (0.0232) p=0.250	+0.0077 (0.0224) p=0.732	+0.0038 (0.0223) p=0.866
Team avg. doubles rank	−0.0003 ^{***} (0.0001) p=0.000	−0.0003 ^{***} (0.0001) p=0.000	−0.0003 ^{***} (0.0001) p=0.000
Opponent avg. doubles rank	+0.0003 ^{***} (0.0001) p=0.000	+0.0003 ^{***} (0.0001) p=0.000	+0.0003 ^{***} (0.0001) p=0.000
Top-100 singles player	−0.0420 (0.0446) p=0.346	−0.0400 (0.0446) p=0.370	−0.0399 (0.0446) p=0.371
Team avg GS appearances	+0.0020 (0.0068) p=0.772	+0.0012 (0.0068) p=0.855	+0.0011 (0.0068) p=0.867
GS experience squared	+0.0000 (0.0003) p=0.953	+0.0000 (0.0003) p=0.905	+0.0000 (0.0003) p=0.897
Pseudo-R ²	0.014	0.014	0.014

Unit = one team per tiebreak (2 obs per tiebreak). Includes set-1 7-pt (463), set-2 7-pt (470), set-3 7-pt (211), and set-3 10-pt match TB (4). Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq (same as Table 3). FE: tournament×year + round. SE clustered by match. Culture AMEs are small and insignificant across all three measures — homophily advantage does not extend to within-match tiebreak outcomes. Ranking controls remain strongly significant ($p < 0.001$), confirming that the model captures real quality differences; experience controls are not significant in the tiebreak sample.

6. Heterogeneity Analysis

Three sets of interactions are estimated on the main match-win sample ($N = 3,744$ team-obs): (a) culture × GS experience, testing whether experienced teams benefit more from cultural alignment; (b) culture × surface, testing surface-specific coordination demands; and (c) culture × Hofstede individualism score (team-level, demeaned), testing whether collectivist teams gain more from cultural homophily.

Hofstede individualism scores (IDV) are taken from Geert Hofstede's official 6-dimension dataset (2015 release, geerthofstede.com). Countries not covered by Hofstede are assigned proxy scores following two tiers: *Tier 1* (direct cultural/linguistic kin, high confidence): BIH→SRB, CYP→GRC, MCO→FRA, MDA→ROU; *Tier 2* (same-region / closest available, moderate confidence): BLR→RUS, BOL→PER, DOM→COL, LBN→Arab composite (IDV=38), TUN→MAR, GEO→TUR, KAZ→RUS, UKR→RUS, UZB→RUS. The team-level score is $IC_team = (IDV_p1 + IDV_p2) / 2$, demeaned across the GS regression sample (mean = 62.99, SD = 20.87).

Table 6a — Heterogeneity: Culture × GS Experience | N = 3,744

Main effect of culture evaluated at zero GS experience; interaction captures how the marginal effect changes per additional Grand Slam appearance.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0351	−0.0275	−0.0306
	(0.0407) p=0.389	(0.0414) p=0.507	(0.0414) p=0.460
C × exp_mean	+0.0001	+0.0060 *	+0.0060 *
	(0.0032) p=0.963	(0.0031) p=0.054	(0.0031) p=0.055
Pseudo-R ²	0.096	0.097	0.097
N	3,744	3,744	3,744

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq. FE: tournament×year + round. SE clustered by match. The negative main effect of culture in cols (2)–(3) is evaluated at zero GS experience (teams with no prior GS appearances); the positive and marginally significant interaction (p≈0.054–0.055) implies the language/proximity advantage grows with experience: at the sample mean of exp_mean ≈ 12 appearances, the implied total marginal effect is approximately +0.044–0.047, consistent with Table 3. Nationality homophily shows no significant experience gradient (p=0.963).

Table 6b — Heterogeneity: Culture × Surface | N = 3,744

Hardcourt (Australian Open + US Open) is the omitted baseline. Clay = Roland Garros; Grass = Wimbledon. Note: since surface is perfectly determined by tournament (each GS is on one surface), the surface main effects may be absorbed by the tournament×year FE; the interaction terms capture whether the culture coefficient differs across surfaces.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — hardcourt baseline	+0.0404 *	+0.0465 **	+0.0404 *
	(0.0230) p=0.080	(0.0222) p=0.036	(0.0221) p=0.068
C × clay	−0.0029	−0.0075	+0.0054
	(0.0376) p=0.938	(0.0368) p=0.838	(0.0370) p=0.884
C × grass	−0.0121	+0.0063	+0.0036
	(0.0404) p=0.765	(0.0389) p=0.871	(0.0389) p=0.925
Pseudo-R ²	0.096	0.096	0.096
N	3,744	3,744	3,744

All surface interactions insignificant (p > 0.76 throughout). The culture advantage documented in Table 3 does not vary significantly across playing surfaces.

Table 6c — Heterogeneity: Culture × Hofstede Individualism (demeaned) | N = 3,744

IC_team_dm = team-average Hofstede IDV score minus the sample mean (62.99). Positive values = more individualist than average. Hypothesis: more individualist teams rely less on in-group bonding → negative θ . Two specifications per culture measure are shown: Spec 1 excludes the main IC effect; Spec 2 includes it.

Variable	Same Nationality		Same Language		Lang. Proximity	
	Spec 1	Spec 2	Spec 1	Spec 2	Spec 1	Spec 2
Culture (C)	+0.0341** (0.0165) p=0.039	+0.0297* (0.0166) p=0.074	+0.0433*** (0.0157) p=0.006	+0.0386** (0.0162) p=0.018	+0.0398** (0.0157) p=0.011	+0.0351** (0.0162) p=0.031
IC_team_dm	—	+0.0012** (0.0006) p=0.045	—	+0.0008 (0.0008) p=0.272	—	+0.0008 (0.0008) p=0.275
C × IC_team_dm (θ)	+0.00060 (0.00053) p=0.258	−0.00054 (0.00079) p=0.490	+0.00079 (0.00046) p=0.087	−0.00004 (0.00089) p=0.960	+0.00082 (0.00046) p=0.076	−0.00001 (0.00089) p=0.990
Pseudo-R ²	0.096	0.097	0.097	0.097	0.097	0.097
N	3,744		3,744		3,744	

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq. FE: tournament×year + round. SE clustered by match. The culture main effects in Spec 1 are consistent with Table 3. The interaction θ is small and insignificant across all specifications ($p > 0.07$), providing no evidence that individualism moderates the cultural homophily advantage. In Spec 2 the main IC_team_dm coefficient is positive and significant for same nationality ($p=0.045$) — teams with higher individualism scores perform better overall — but is insignificant for the language measures ($p>0.27$). The Spec 1 interaction terms for language and proximity are marginally significant ($p=0.087$ and $p=0.076$), suggesting tentative evidence that more individualist teams benefit more from language similarity.